Research proposal

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1. Problem and Gap

There has been a change in demand for certain occupations due to the spread of artificial intelligence usage (Ando et al. 2026). This makes understanding how affected workers can find new career paths increasingly important. Based on (Neffke et al. 2017a), skill-relatedness could be very useful in identifying possible mobility paths in relevant cases. Even though (Cazzaniga et al. 2025) have addressed this issue, they did not examine it through skill-relatedness and more importantly, their data only reaches till 2019. As such, it does not cover the recent expansion of generative AI usage. We intend to contribute to filling this gap by examining how worker mobility can be described through skill-relatedness, based on recent data from Germany (Statistik der Bundesagentur für Arbeit, n.d.).

2. Research question and hypothesis

We were interested to see what skill-relatedness can tell us about mobility from occupations that are the most affected by AI. More specifically: Where are the most AI-exposed occupations found in the skill-relatedness network and are there any, less-exposed alternatives near them in terms of required skills that seem to offer a transitional opportunity? Have transitions from the examined occupations become more (or less) common recently? Are the alternative occupations in reach more, less, or similarly AI-affected in general than the starting occupation?

We will use skill-relatedness matrices building on Neffke et al. (2017a). This network-based approach gives us an opportunity to identify and evaluate alternatives effectively. According to our hypotheses, such transitions could have become more frequent in the past several years, however, the closest alternatives might still be similarly AI-exposed.

3. Data and method

We will identify the most affected occupations using three data sources. First, ILO generative-AI index (Gmyrek et al. 2025): ranks each ISCO-08 occupation (4-digit) on how exposed it is to generative AI. Second, Skill-relatedness (Neffke et al., 2017b): a matrix of 329 German occupations (KldB88, 3-digit), based on worker flows between 1975 and 2010. Last, BA mobility tables (Bundesagentur für Arbeit 2019-2025): job starts between 36 occupation groups (KldB 2010) for 2025. Following Neffke et al. (2017b), we will build a graph from the 329 German occupations, where the occupations are nodes and directed links from occupation i to occupation j are with weight $w_{ij} = \max(SR_{ij}, 0)$. Exposure comes from the ILO genAI index (Gmyrek et al. 2025). To match the sources we will create a crosswalk with them. For each exposed job i, we will compute:

Escape Share (E): the fraction of the link weight that leads to low-exposure occupations (we define it as the lower half).

$$E_i = \frac{\sum_{j \in S} w_{ij}}{\sum_{j \in C_i} w_{ij}}$$

Exposure Gap (i): Computed for its 10 strongest neighbours (Ci), where a positive gap shows that the nearest alternatives are less exposed.

$$\Delta_i = e_i - \frac{\sum_{j \in C_i} w_{ij} e_j}{\sum_{j \in C_i} w_{ij}}$$

To test if transitions have become more frequent, we will use the Bundesagentur für Arbeit mobility tables from 2019 to 2025. For each group we will compute the outflow rate r and compare the change in exposed groups with the change in other groups.
Data access and Plan B. We already have the IAB matrices, the BA mobility tables, and the ILO scores are public. Our main risk is the crosswalk between KldB88 and ISCO-08, which has not been done yet. If it fails, the Plan B is to build the network from either the 120-group occ_SUF file or the 36 BA groups.

4. Expected results and impact

Our research can help identify realistic alternatives to AI-exposed occupations. This can help workers in choosing new potential paths. More importantly, the findings could support policymakers in charge of retraining programmes to identify both affected areas and relevant alternative paths that would make a useful base for what training opportunities are needed the most. Such retraining programmes, especially in terms of AI, are already underway in Germany (OECD 2026), among other countries.

Bibliography

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